

2. Specification

Model			VMV-S121AREHDA3	VMV-S140AREHDA3	VMV-S155AREHDA3
Power supply		Ph/V/Hz	3/380~415/50/60		
Cooling	Rated capacity	kW	12.60	14.00	15.50
	Rated capacity	kBtu/h	43.0	47.8	52.9
	Rated power input	kW	3.11	3.51	4.31
	Max. power input	kW	7.2	7.5	7.8
	EER		4.05	3.99	3.60
	Rated current	A	4.9	5.6	6.8
	Max. current	A	11.4	11.9	12.3
Heating	Rated capacity	kW	14.2	16.0	18.0
	Rated capacity	kBtu/h	48.5	54.6	61.4
	Rated power input	kW	3.2	3.7	4.4
	Max. power input	kW	6.9	7.2	7.5
	COP		4.47	4.30	4.10
	Rated current	A	5.0	5.9	6.9
	Max. current	A	10.9	11.4	11.9
Compressor	Brand		MITSUBISHI ELECTRIC		
	Model		MNB42FFDMC-L		
	Type		Rotary	Rotary	Rotary
	Compressor quantity		1 INV	1 INV	1 INV
	Capacity	W	13780	13780	13780
	Power Input	W	4060	4060	4060
	Rated current(RLA)	A	12.3	12.3	12.3
	Speed	rps	75	80	85
	Crankcase Heater	W	28	28	28
	Refrigerant oil brand		Itochu.,LTD.,Shanghai	Itochu.,LTD.,Shanghai	Itochu.,LTD.,Shanghai
	Refrigerant oil type		FV50S	FV50S	FV50S
	Refrigerant oil charge	ml	1400	1400	1400
Outdoor fan motor	Brand		BROAD OCEAN	BROAD OCEAN	BROAD OCEAN
	Model		SIC-88FWJ-F1180-1	SIC-88FWJ-F1180-1	SIC-88FWJ-F1180-1
	Voltage		310V	310V	310V
	IP Class		IP44	IP44	IP44
	Type		DC	DC	DC
	Insulation class		E	E	E
	Safe class		I	I	I
	Power Input	W	225*2	225*2	225*2
	Output	W	180*2	180*2	180*2
	Rated current	A	0.40	0.40	0.40
	Capacitor	μF	/	/	/
	Speed	rpm	900	900	900
Outdoor fan	Brand		GUOEN	GUOEN	GUOEN
	Model		/	/	/
	Material		Plastic	Plastic	Plastic
	Type		Axial	Axial	Axial
	Diameter	mm	550	550	550
	Height	mm	200	200	200

Part name	Sign	Function	Date	Note
Compressor	/	Capacity control, meet indoor load request by adjusting frequency and opening and closing fixing frequency compressor.		20℃
Pressure switch	HPS	High pressure protection	4.15Mpa, OFF	
Electronic expansion valve	EEV	In heating, refrigerant flow control (subcooling valve)	ū 3.0	
Solenoid valve	SV1	1. Keep balance of high/low pressure when compressor starts up and stops 2. High/low pressure protection	AC220V Open when power is on, close when power is off.	2A
4-way valve	4WV	Changing over between cooling and heating	AC220V electrified in heating; powered off in cooling or defrosting.	
Pressure sensor	PD	In heating, compressor frequency adjustment, abnormal pressure protection		
	PS	In cooling, compressor frequency adjustment, abnormal pressure protection		
Temp. sensor	TD	Detect the top temp. of compressor	R(80℃)=50K B(25/80℃)=4450K	
	TS	Detect the top suction of compressor	R(80℃)=10K B(25/80℃)=3700K	
	TA	Detect ambient temp., set primary fan speed and control defrost condition		
	TX1	Detect the temp. of before and after the supercooling valve to control the supercooling valve open angle.		
	TX2			
	TE(1)	Detect frost condition of outdoor heat exchanger		
Heater	Chi	Used to heat oil in inverter compressor	28W, 220V, one	

- The outdoor gas pipe and the refrigerant distributing pipe, as well the refrigerant distributing pipe and the branch pipe should be welded with hard solder.

Weld the pipe at the same time charge the nitrogen. Or it will cause a number of impurity (a film of oxidation) to clog the capillary and the expansion valve, further cause the deadly failure.	Seal the pipe end with adhesive tape or the stopper to increase the resistance, fill up the pipe with nitrogen.
Protect the pipe end against the water, impurity into the pipes (welding after being flat, or being sealed with adhesive tape).	
The refrigerant pipe should be clean. The nitrogen should flow under the pressure of about 0.2Mpa and when charging the nitrogen, stop up the end of the pipe by hand to enhance the pressure in the pipe, then loose the hand (meanwhile stop up the other end).	
When connecting the pipes, close the valves fully.	
When welding the valve and the pipes, use the wet cloth to cool down the valve and the pipes.	

(6) Leakage test

- The outdoor unit has been executed the leakage test in the factory. After connecting the distributing pipe, execute the leakage test from the outdoor check valve and the indoor. Besides, while testing, the valves should be close.
- Refer to the below figure to charge the nitrogen into the unit to take a test. Never use the chlorin, oxygen, flammable gas in the leakage test. Apply pressure both on the gas pipe and the liquid pipe.
- Apply the pressure step by step to the target pressure.
 - Apply the pressure to 0.5MPa for more than 5 minutes, confirm if pressure goes down.
 - Apply the pressure to 1.5MPa for more than 5 minutes, confirm if pressure goes down.
 - Apply the pressure to the target pressure (4.0MPa), record the temp. and the pressure.
 - Leave it at 4.0MPa for over 1 day, if pressure does not go down, the test is passed. Meanwhile, when the temp. changes for 1degree, pressure will change 0.01MPa as well. Correct the pressure.
 - After confirmation of a~d, if pressure goes down, there is leakage. Check the brazing position, flared position by laying on the soap. modify the leakage point and take another leakage test.
- After leakage test, must execute the evacuation.

(7) Evacuation

Evacuate at the check valve of liquid stop valve and both sides of the gas stop valve.

Operation procedure:

